



WELL-POSEDNESS ANALYSIS FOR A FLUID-STRUCTURE INTERACTION PROBLEM FORMULATED VIA THE FICTITIOUS DOMAIN METHOD

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ABSTRACT. This paper develops a generalized framework for Fluid–Structure Interaction (FSI) problems, focusing on the formulation of generalized interface conditions inspired by the fictitious domain methodology, which enables the treatment of a wide range of physical configurations. The weak formulation is established within a product functional space, providing a unified setting for analyzing the coupled fluid and solid subsystems. The principal theoretical contribution concerns the existence of solutions, which is demonstrated using tools from semigroup theory and abstract nonlinear evolution equations, while truncation techniques are employed to manage the nonlinear terms. In the two-dimensional case, uniqueness is further established through energy estimates and analytical properties of the associated trilinear form.

1. Introduction. The mathematical and computational study of FSI has advanced significantly, focusing on both theoretical foundations and numerical methods. Classical FSI formulations typically impose standard interface conditions, such as no-slip velocity continuity and stress balance. Rigorous studies have established the existence of weak solutions for incompressible, viscous flows coupled with deformable structures [2, 10, 12], as well as local existence of strong solutions for nonlinear FSI systems [6, 7, 9]. Fictitious domain methods with sharp interfaces have been introduced to handle complex interface geometries and discontinuities in material properties [1, 5]. These contributions provide a solid theoretical framework that supports both the existence and uniqueness of solutions for a wide range of FSI configurations [3].

In this manuscript, we develop a generalized FSI framework that incorporates parameterized, fictitious domain-type interface conditions. By embedding the solid structure within an extended fluid domain, the coupling terms are reformulated to handle more general phenomena, including partial slip and stress discontinuities. The weak formulation integrates these generalized conditions, and theoretical results on the existence and uniqueness of solutions are established using semigroup theory

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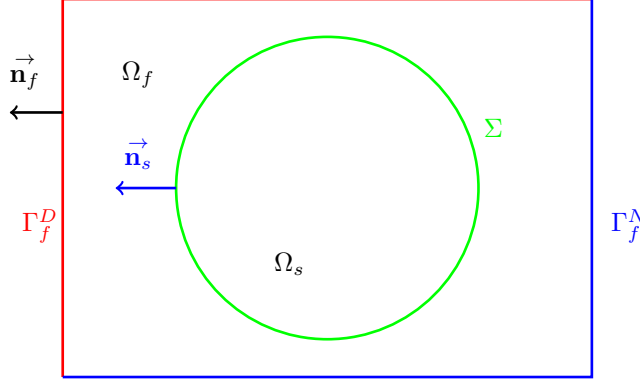


FIGURE 1. An example of an elastic domain Ω_s embedded in a fluid domain Ω_f

[2] and functional analysis. This approach extends beyond classical assumptions, offering a unified model suitable for complex fluid-structure interactions.

The paper is organized as follows. Section 2 introduces the governing equations, initial and boundary conditions, and the functional framework. The weak formulation is developed in Section 3. Proofs of existence along with uniqueness for $d = 2$ are detailed in Section 4.

2. Fluid-Structure Interaction Model. We consider a fluid-structure interaction (FSI) problem, where a deformable structure is immersed in a fluid domain. The fluid domain is denoted by $\Omega_f \subset \mathbb{R}^d$ (typically $d = 2$ or $d = 3$), and the structure domain is denoted by $\Omega_s \subset \mathbb{R}^d$. The interaction between the fluid and the structure occurs along the interface $\Sigma \subset \mathbb{R}^{d-1}$. The outer boundary of the fluid domain is denoted by $\Gamma_f = \Gamma_f^D \cup \Gamma_f^N$, where Γ_f^D represents the Dirichlet part and Γ_f^N represents the Neumann part of the boundary.

2.1. Governing equations, initial and boundary conditions. The fluid flow is described by the *incompressible Navier–Stokes equations*. Let v^f and p denote the velocity field and pressure of the fluid, respectively. The governing equations are:

- The momentum equation (Newton’s second law):

$$\frac{\partial v^f}{\partial t} + (v^f \cdot \nabla)v^f - \nabla \cdot \sigma_f(v^f, p) = \mathbf{f}^f, \quad \text{in } (0, T) \times \Omega_f.$$

- The continuity equation (mass conservation):

$$\nabla \cdot v^f = 0, \quad \text{in } (0, T) \times \Omega_f,$$

where the fluid stress tensor $\sigma_f(v^f, p)$ is given by

$$\sigma_f(v^f, p) = -p\mathbf{I} + 2\mu_f \varepsilon(v^f),$$

with $\varepsilon(v^f) = \frac{1}{2} (\nabla v^f + (\nabla v^f)^\top)$ being the rate-of-strain tensor (i.e., the symmetric part of the velocity gradient), μ_f the dynamic viscosity of the fluid, and $\mathbf{f}^f \in L^2(0, T; L^2(\Omega_f)^d)$ is a volume force acting on the fluid.

The structure is governed by the *linear elastodynamics equation*. Let u^s denote the displacement field of the structure. Then,

$$\frac{\partial^2 u^s}{\partial t^2} - \nabla \cdot \sigma_s(u^s) = \mathbf{f}^s, \quad \text{in } (0, T) \times \Omega_s, \quad (2.1)$$

where $\mathbf{f}^s \in H^1(0, T; L^2(\Omega_s)^d)$ is a volume force acting on the structure, and the stress tensor σ_s is defined by

$$\sigma_s(u^s) = 2\mu \varepsilon(u^s) + \lambda \operatorname{tr}(\varepsilon(u^s))I, \quad \text{with} \quad \varepsilon(u^s) = \frac{1}{2} (\nabla u^s + \nabla u^{s\top}),$$

and μ_s and λ are the Lamé coefficients related to the Young's modulus E and Poisson's ratio ν , by

$$\mu_s = \frac{E}{2(1+\nu)}, \quad \lambda_s = \frac{E\nu}{(1+\nu)(1-2\nu)}.$$

The boundary conditions are specified for the fluid velocity and stress as follows:

$$\begin{aligned} v^f &= \mathbf{0}, \quad \text{on } (0, T) \times \Gamma_f^D, \\ \sigma_f(v^f, p) \cdot \mathbf{n}_N^f &= \mathbf{g}_N^f, \quad \text{on } (0, T) \times \Gamma_f^N, \end{aligned}$$

where $\mathbf{g}_N^f \in L^2(0, T; H^{1/2}(\Gamma_f^N)^d)$ is a prescribed traction vector, and $\mathbf{n}_N^f$ is the unit outward normal to Γ_f^N with respect to the fluid domain Ω_f .

Initial conditions for both the fluid and the structure are given by:

$$\begin{aligned} v^f|_{t=0} &= v_0^f, \quad \text{in } \Omega_f, \\ u^s|_{t=0} &= u_0^s, \quad \frac{\partial u^s}{\partial t}\bigg|_{t=0} = v_0^s, \quad \text{in } \Omega_s, \end{aligned}$$

where $v^f \in H^1(\Omega_f)^d$ and $u_0^s, v_0^s \in H^1(\Omega_s)^d$.

2.2. Interface conditions. The interface conditions on $(0, T) \times \Sigma$ couple the fluid and the structure through the balance of stresses and velocities across the interface. They are given by:

$$\begin{aligned} \llbracket \sigma \cdot \mathbf{n} \rrbracket_\Sigma &\equiv \sigma_f \cdot \mathbf{n} - \sigma_s \cdot \mathbf{n} = \mathbf{M} \bar{v}|_\Sigma - \mathbf{h}, \quad \text{on } (0, T) \times \Sigma, \\ \overline{\sigma \cdot \mathbf{n}}|_\Sigma &\equiv \frac{1}{2} (\sigma_f \cdot \mathbf{n} + \sigma_s \cdot \mathbf{n}) = \mathbf{S} \llbracket v \rrbracket_\Sigma - \mathbf{g}, \quad \text{on } (0, T) \times \Sigma, \end{aligned}$$

where

$$\bar{v}|_\Sigma \equiv \frac{1}{2} \left(v|_\Sigma^f + \frac{\partial u^s}{\partial t}\bigg|_\Sigma \right), \quad \llbracket v \rrbracket_\Sigma \equiv v|_\Sigma^f - \frac{\partial u^s}{\partial t}\bigg|_\Sigma,$$

and $\mathbf{n} = \mathbf{n}_s$ denotes the unit outward normal to Σ with respect to the structure domain Ω_s . The functions $\mathbf{g}, \mathbf{h} \in L^2(0, T; H^{-1/2}(\Sigma)^d)$ represent interface force terms. The operators $\mathbf{S}, \mathbf{M} \in \mathbb{L}^\infty(\Sigma)^{d \times d}$ are measurable, bounded matrix-valued functions that satisfy the following ellipticity assumption:

$$\forall \xi \in \mathbb{R}^d, \quad \xi^\top \mathbf{M}(\mathbf{x}) \xi > 0, \quad \xi^\top \mathbf{S}(\mathbf{x}) \xi > 0, \quad \text{for a.e. } \mathbf{x} \in \Sigma. \quad (2.2)$$

Remark 2.1. For brevity, we omit the subscript Σ in the notation for both the jump and the mean value across the interface. That is,

$$\llbracket v \rrbracket \equiv \llbracket v \rrbracket_\Sigma, \quad \bar{v} \equiv \bar{v}_\Sigma, \quad \llbracket \sigma \cdot \mathbf{n} \rrbracket \equiv \llbracket \sigma \cdot \mathbf{n} \rrbracket_\Sigma, \quad \overline{\sigma \cdot \mathbf{n}} \equiv \overline{\sigma \cdot \mathbf{n}}_\Sigma.$$

We now summarize the full continuous model governing the fluid-structure interaction (FSI) problem which is given by (2.3) below:

$$\frac{\partial v^f}{\partial t} + (v^f \cdot \nabla)v^f - \nabla \cdot \boldsymbol{\sigma}_f(v^f, p) = \mathbf{f}^f \quad \text{in } (0, T) \times \Omega_f, \quad (2.3a)$$

$$\nabla \cdot v^f = 0 \quad \text{in } (0, T) \times \Omega_f, \quad (2.3b)$$

$$\frac{\partial^2 u^s}{\partial t^2} - \nabla \cdot \boldsymbol{\sigma}_s(u^s) = \mathbf{f}^s \quad \text{in } (0, T) \times \Omega_s, \quad (2.3c)$$

$$v^f = 0 \quad \text{on } (0, T) \times \Gamma_f^D, \quad (2.3d)$$

$$\boldsymbol{\sigma}_f(v^f, p) \cdot \mathbf{n} = \mathbf{g}_N^f \quad \text{on } (0, T) \times \Gamma_f^N, \quad (2.3e)$$

$$\llbracket \boldsymbol{\sigma} \cdot \mathbf{n} \rrbracket = \mathbf{M} \bar{v} - \mathbf{h}, \quad \text{on } (0, T) \times \Sigma, \quad (2.3f)$$

$$\overline{\boldsymbol{\sigma} \cdot \mathbf{n}} = \mathbf{S} \llbracket v \rrbracket - \mathbf{g}, \quad \text{on } (0, T) \times \Sigma, \quad (2.3g)$$

$$v^f(0, \cdot) = v_0 \quad \text{in } \Omega_f, \quad (2.3h)$$

$$u^s(0, \cdot) = u_0, \quad v^s(0, \cdot) = u_1 \quad \text{in } \Omega_s. \quad (2.3i)$$

To gain more insight into the stress distribution at the fluid-structure interface, we express the normal stresses $\boldsymbol{\sigma}_f \cdot \mathbf{n}$ and $\boldsymbol{\sigma}_s \cdot \mathbf{n}$ in terms of the fluid and structure velocities v^f and v^s . By rearranging the interface conditions (2.3f)–(2.3g), we decompose the stress contributions into terms depending on v^f , v^s , and the interface data $\mathbf{g}$ and $\mathbf{h}$. The resulting expressions for the normal stresses are given by:

$$\begin{cases} \boldsymbol{\sigma}_s \cdot \mathbf{n} = \left(\mathbf{S} - \frac{1}{4} \mathbf{M} \right) v^f - \left(\mathbf{S} + \frac{1}{4} \mathbf{M} \right) v^s - \mathbf{g} + \frac{1}{2} \mathbf{h}, & \text{on } (0, T) \times \Sigma, \\ \boldsymbol{\sigma}_f \cdot \mathbf{n} = \left(\mathbf{S} + \frac{1}{4} \mathbf{M} \right) v^f - \left(\mathbf{S} - \frac{1}{4} \mathbf{M} \right) v^s - \mathbf{g} - \frac{1}{2} \mathbf{h}, & \text{on } (0, T) \times \Sigma. \end{cases} \quad (2.4)$$

Having established the generalized interface conditions (2.3f)–(2.3g), we now turn our attention to specific, simplified cases that can be derived as limiting forms of this general framework. These classical models are widely encountered in the literature on fluid-structure interaction and serve as special cases obtained through particular choices of the interface parameters. By appropriately selecting the coefficients $\mathbf{M}$, $\mathbf{S}$, $\mathbf{g}$, $\mathbf{h}$, we recover several well-known configurations that offer important insight into physical applications. For that, let $\varepsilon > 0$ be a real penalty parameter, and let $\mathbf{I}$ denote the identity matrix. We present the following cases:

• **Case 1 (No-slip with traction continuity):**

Assume $\mathbf{h} = \mathbf{g} = \mathbf{0}$, $\mathbf{M} = \varepsilon \mathbf{I}$, and $\mathbf{S} = \frac{1}{\varepsilon} \mathbf{I}$. The interface conditions become

$$\begin{cases} \llbracket \boldsymbol{\sigma} \cdot \mathbf{n} \rrbracket = \varepsilon \bar{v}, \\ \varepsilon \overline{\boldsymbol{\sigma} \cdot \mathbf{n}} = \llbracket v \rrbracket. \end{cases}$$

In the limit as $\varepsilon \rightarrow 0$, we obtain the classical interface conditions:

$$\begin{cases} \llbracket v \rrbracket = \mathbf{0}, \\ \llbracket \boldsymbol{\sigma} \cdot \mathbf{n} \rrbracket = \mathbf{0}. \end{cases}$$

• **Case 2 (Velocity jump with traction continuity):**

Assume $\mathbf{h} = \mathbf{0}$, $\mathbf{g} = \frac{1}{\varepsilon} \mathbf{q}$, $\mathbf{M} = \varepsilon \mathbf{I}$, and $\mathbf{S} = \frac{1}{\varepsilon} \mathbf{I}$. In the limit $\varepsilon \rightarrow 0$, the

interface conditions reduce to:

$$\begin{cases} \llbracket v \rrbracket = \mathbf{q}, \\ \llbracket \boldsymbol{\sigma} \cdot \mathbf{n} \rrbracket = \mathbf{0}. \end{cases}$$

• **Case 3 (No-slip with traction jump):**

Assume $\mathbf{h} \neq \mathbf{0}$, $\mathbf{g} = \mathbf{0}$, $\mathbf{M} = \varepsilon \mathbf{I}$, and $\mathbf{S} = \frac{1}{\varepsilon} \mathbf{I}$. Then, as $\varepsilon \rightarrow 0$, the limiting interface conditions become:

$$\begin{cases} \llbracket v \rrbracket = \mathbf{0}, \\ \llbracket \boldsymbol{\sigma} \cdot \mathbf{n} \rrbracket = \mathbf{h}. \end{cases}$$

• **Case 4 (Velocity and traction jumps):**

This case combines the previous two. Let $\mathbf{g} = \frac{1}{\varepsilon} \mathbf{q}$, $\mathbf{h} \neq \mathbf{0}$, $\mathbf{M} = \varepsilon \mathbf{I}$, and $\mathbf{S} = \frac{1}{\varepsilon} \mathbf{I}$. Then, as $\varepsilon \rightarrow 0$, we obtain:

$$\begin{cases} \llbracket v \rrbracket = \mathbf{q}, \\ \llbracket \boldsymbol{\sigma} \cdot \mathbf{n} \rrbracket = \mathbf{h}. \end{cases}$$

All cases are summarized in Table 1.

Case #	Interface Conditions	Parameters on Σ
1	No-slip and stress continuity	$\mathbf{h} = \mathbf{g} = \mathbf{0}$, $\mathbf{M} = \varepsilon \mathbf{I}$, $\mathbf{S} = \frac{1}{\varepsilon} \mathbf{I}$
2	Slip and stress continuity	$\mathbf{h} = \mathbf{0}$, $\mathbf{g} = \frac{1}{\varepsilon} \mathbf{q}$, $\mathbf{M} = \varepsilon \mathbf{I}$, $\mathbf{S} = \frac{1}{\varepsilon} \mathbf{I}$
3	No-slip and stress discontinuity	$\mathbf{h} \neq \mathbf{0}$, $\mathbf{g} = \mathbf{0}$, $\mathbf{M} = \varepsilon \mathbf{I}$, $\mathbf{S} = \frac{1}{\varepsilon} \mathbf{I}$
4	Slip and stress discontinuity	$\mathbf{h} \neq \mathbf{0}$, $\mathbf{g} = \frac{1}{\varepsilon} \mathbf{q}$, $\mathbf{M} = \varepsilon \mathbf{I}$, $\mathbf{S} = \frac{1}{\varepsilon} \mathbf{I}$

TABLE 1. Variants of interface conditions for fluid–structure interaction problems, based on the limiting behavior as $\varepsilon \rightarrow 0$.

2.3. Functional Spaces, Inner Products and Norms. We consider the following functional spaces for the fluid and structure problems. The fluid velocity v^f is sought in the Sobolev space

$$V_f \equiv \{v^f \in H^1(\Omega_f)^d : \nabla \cdot v^f = 0 \text{ in } \Omega_f, \ v^f = \mathbf{0} \text{ on } \Gamma_f^D\},$$

while the fluid pressure p belongs to the space $L_0^2(\Omega_f)$ defined by

$$L_0^2(\Omega_f) \equiv \left\{ p \in L^2(\Omega_f) : \int_{\Omega_f} p = 0 \right\}.$$

The structure displacement u^s is sought in

$$V_s \equiv H^1(\Omega_s)^d.$$

For any $v, w \in V_f$, the space V_f is equipped with the inner product and associated norm:

$$(v, w)_{1,f} \equiv \int_{\Omega_f} \varepsilon(v(\mathbf{x})) : \varepsilon(w(\mathbf{x})) \, d\mathbf{x}, \quad \|v\|_{1,f} = \left(\int_{\Omega_f} |\varepsilon(v(\mathbf{x}))|^2 \, d\mathbf{x} \right)^{1/2},$$

where $\varepsilon(v) = \frac{1}{2}(\nabla v + \nabla v^\top)$ is the symmetric part of the gradient.

The inner product of V_s is defined for every $v, w \in V_s$ by:

$$(u, v)_{1,s} \equiv \int_{\Omega_s} u(\mathbf{x}) \cdot v(\mathbf{x}) d\mathbf{x} + \int_{\Omega_s} \boldsymbol{\sigma}_s(u(\mathbf{x})) : \varepsilon(v(\mathbf{x})) d\mathbf{x},$$

and the associated norm is given by:

$$\|u\|_{1,s} = \left(\|u\|_{0,s}^2 + \int_{\Omega_s} \boldsymbol{\sigma}_s(u) : \varepsilon(u) d\mathbf{x} \right)^{1/2}.$$

Let V_f' and V_s' be the duals of V_f and V_s , respectively. For any $\ell_f \in V_f'$ and $\ell_s \in V_s'$, the dual norms are defined as:

$$\|\ell_f\|_{V_f'} := \sup_{\varphi \in V_f, \|\varphi\|_{1,f}=1} |\langle \ell_f, \varphi \rangle_{V_f', V_f}|, \quad \text{and} \quad \|\ell_s\|_{V_s'} := \sup_{\varphi \in V_s, \|\varphi\|_{1,s}=1} |\langle \ell_s, \varphi \rangle_{V_s', V_s}|,$$

where $\langle \cdot, \cdot \rangle_{V_\omega', V_\omega}$ is the duality product between V_ω' and V_ω for $\omega \in \{f, s\}$.

For any $v, w \in L^2(\Omega_\omega)^d$ with $\omega \in \{f, s\}$, the standard L^2 inner product and norm are given by:

$$(v, w)_{0,\omega} \equiv \int_{\Omega_\omega} v \cdot w d\mathbf{x}, \quad \|v\|_{0,\omega}^2 = (v, v)_{0,\omega}.$$

We also denote by $\langle \cdot, \cdot \rangle_\Sigma \equiv \langle \cdot, \cdot \rangle_{H^{-1/2}, H^{1/2}}$ the duality pairing between the Sobolev space $H^{-1/2}(\Sigma)$ and $H^{1/2}(\Sigma)$.

We define the fluid and structure energy spaces by

$$H_s := [L^2(\Omega_s)]^d,$$

and

$$H_f := \{v^f \in [L^2(\Omega_f)]^d; \nabla \cdot v^f = 0 \text{ in } \mathcal{D}'(\Omega_f), \quad v^f \cdot \mathbf{n} = 0 \text{ on } \Gamma_f^D\}.$$

Equivalently, H_f is the closure of V_f in $[L^2(\Omega_f)]^d$:

$$H_f = \overline{V_f}^{[L^2(\Omega_f)]^d}.$$

Equipped with the usual L^2 -inner products, the spaces H_f and H_s are Hilbert spaces, and we have the continuous and dense embeddings

$$V_f \hookrightarrow H_f \hookrightarrow V_f', \quad V_s \hookrightarrow H_s \hookrightarrow V_s'.$$

Finally, we define the product space:

$$V \equiv \{v = (v^f, u^s) \in V_f \times V_s : \llbracket \boldsymbol{\sigma} \cdot \mathbf{n} \rrbracket = \mathbf{M} \bar{v} - \mathbf{h}, \quad \overline{\boldsymbol{\sigma} \cdot \mathbf{n}} = \mathbf{S} \llbracket v \rrbracket - \mathbf{g} \text{ on } \Sigma\}, \quad (2.5)$$

where the jump and average conditions are imposed on the interface Σ , and the vector field v consists of the fluid velocity $v^f \in V_f$ and the structure displacement $u^s \in V_s$.

For any $u, v \in L^2(\Omega_f)^d \times L^2(\Omega_s)^d \equiv \{u = (u^f, u^s) : u^f \in L^2(\Omega_f)^d, \quad u^s \in L^2(\Omega_s)^d\}$, we define the standard scalar product and norm:

$$(u, v) \equiv (u^f, v^f)_{0,f} + (u^s, v^s)_{0,s}, \quad \|u\|^2 \equiv \|u^f\|_{0,f}^2 + \|u^s\|_{0,s}^2. \quad (2.6)$$

For any $u, v \in V$, we define the corresponding H^1 -based scalar product and norm by:

$$(u, v)_1 \equiv (u^f, v^f)_{1,f} + (u^s, v^s)_{1,s}, \quad \|u\|_1^2 \equiv \|u^f\|_{1,f}^2 + \|u^s\|_{1,s}^2. \quad (2.7)$$

For time-dependent functions, we use the standard Bochner spaces:

$$L^2(0, T; X), \quad L^\infty(0, T; X), \quad H^1(0, T; X),$$

where X may be any of the previously defined spaces such as H_f, V_f, V_s, V , or $L^2(\Omega_\omega)$ for $\omega \in \{f, s\}$.

3. Weak Formulation. We now establish the weak formulation of the fluid-structure interaction problem. Let $\varphi^f \in V_f$, $q^f \in L^2(\Omega_f)$, and $\varphi^s \in V_s$ be test functions corresponding to equations (2.3a), (2.3b), and (2.3c), respectively. Multiplying the fluid momentum equation (2.3a) by φ^f and integrating over Ω_f , we obtain:

$$\int_{\Omega_f} \left(\frac{\partial v^f}{\partial t} + (v^f \cdot \nabla) v^f \right) \cdot \varphi^f d\mathbf{x} - \int_{\Omega_f} \nabla \cdot \boldsymbol{\sigma}_f(v^f, p) \cdot \varphi^f d\mathbf{x} = \int_{\Omega_f} \mathbf{f}^f \cdot \varphi^f d\mathbf{x}.$$

Applying Green's formula to the second term on the left-hand side and incorporating the boundary contributions on Γ_f^N and the fluid-structure interface Σ , we obtain the weak form of the fluid momentum equation:

$$\begin{aligned} \left(\frac{\partial v^f}{\partial t}, \varphi^f \right)_{0,f} + ((v^f \cdot \nabla) v^f, \varphi^f)_{0,f} + (\boldsymbol{\sigma}_f(v^f, p), \nabla \varphi^f)_{0,f} - \langle \boldsymbol{\sigma}_f(v^f, p) \cdot \mathbf{n}_f, \varphi^f \rangle_\Sigma \\ = (\mathbf{f}^f, \varphi^f)_{0,f} + \langle \mathbf{g}_N^f, \varphi^f \rangle_{\Gamma_f^N}, \quad \forall \varphi^f \in V_f. \end{aligned} \quad (3.1)$$

The incompressibility constraint (2.3b) is enforced in weak form as:

$$\int_{\Omega_f} q^f \nabla \cdot v^f d\mathbf{x} = 0, \quad \forall q^f \in L^2(\Omega_f). \quad (3.2)$$

For the structure, multiplying (2.3c) by $\varphi^s \in V_s$ and integrating over Ω_s , we obtain:

$$\int_{\Omega_s} \frac{\partial^2 u^s}{\partial t^2} \cdot \varphi^s d\mathbf{x} - \int_{\Omega_s} \nabla \cdot \boldsymbol{\sigma}_s(u^s) \cdot \varphi^s d\mathbf{x} = \int_{\Omega_s} \mathbf{f}^s \cdot \varphi^s d\mathbf{x}.$$

Using Green's formula, the weak form becomes:

$$\left(\frac{\partial^2 u^s}{\partial t^2}, \varphi^s \right)_{0,s} + (\boldsymbol{\sigma}_s(u^s), \nabla \varphi^s)_{0,s} - \langle \boldsymbol{\sigma}_s \cdot \mathbf{n}_s, \varphi^s \rangle_\Sigma = (\mathbf{f}^s, \varphi^s)_{0,s}. \quad (3.3)$$

Adding equations (3.1) and (3.3), and substituting $\boldsymbol{\sigma}_f = -p\mathbf{I} + 2\mu_f \varepsilon(v^f)$, we get the combined weak formulation:

$$\begin{aligned} \left(\frac{\partial v^f}{\partial t}, \varphi^f \right)_{0,f} + \left(\frac{\partial^2 u^s}{\partial t^2}, \varphi^s \right)_{0,s} + ((v^f \cdot \nabla) v^f, \varphi^f)_{0,f} + (\varepsilon(v^f), \varepsilon(\varphi^f))_{0,f} \\ + (\boldsymbol{\sigma}_s(u^s), \varepsilon(\varphi^s))_{0,s} - \langle \boldsymbol{\sigma}_f \cdot \mathbf{n}_f, \varphi^f \rangle_\Sigma - \langle \boldsymbol{\sigma}_s \cdot \mathbf{n}_s, \varphi^s \rangle_\Sigma \\ = (\mathbf{f}^f, \varphi^f)_{0,f} + (\mathbf{f}^s, \varphi^s)_{0,s} + \langle \mathbf{g}_N^f, \varphi^f \rangle_{\Gamma_f^N}, \quad \forall \varphi^f \in V_f, \forall \varphi^s \in V_s. \end{aligned} \quad (3.4)$$

Now, applying the identity from [1, 5]

$$\langle u^+, v^+ \rangle_\Sigma - \langle u^-, v^- \rangle_\Sigma = \langle \llbracket u \rrbracket, \bar{v} \rangle_\Sigma + \langle \bar{u}, \llbracket v \rrbracket \rangle_\Sigma, \quad (3.5)$$

and using the fact that $\mathbf{n}_s = -\mathbf{n}_f$, the interface terms in (3.4) become:

$$\begin{aligned} -\langle \boldsymbol{\sigma}_f \cdot \mathbf{n}_f, \varphi^f \rangle_\Sigma - \langle \boldsymbol{\sigma}_s \cdot \mathbf{n}_s, \varphi^s \rangle_\Sigma &= \langle \llbracket \boldsymbol{\sigma} \cdot \mathbf{n}_s \rrbracket, \bar{\varphi} \rangle_\Sigma + \langle \overline{\boldsymbol{\sigma} \cdot \mathbf{n}_s}, \llbracket \varphi \rrbracket \rangle_\Sigma \\ &= \langle \mathbf{M} \bar{v}, \bar{\varphi} \rangle_\Sigma - \langle \mathbf{h}, \bar{\varphi} \rangle_\Sigma + \langle \mathbf{S} \llbracket v \rrbracket, \llbracket \varphi \rrbracket \rangle_\Sigma - \langle \mathbf{g}, \llbracket \varphi \rrbracket \rangle_\Sigma. \end{aligned}$$

Definition 3.1. Let $(v_0^f, u_0^s, v_0^s) \in V_f \times V_s \times V_s$, and $T > 0$. The couple (v^f, u^s) of $L^\infty(0, T; V)$ is a weak solution of problem (2.3) if:

- The initial data is satisfied:

$$(v^f, \partial_t u^s, u^s)|_{t=0} = (v_0^f, v_0^s, u_0^s).$$

- For almost every $t \in (0, T)$, and for all $\varphi^f \in V_f$, $\varphi^s \in V_s$, the following weak form holds:

$$\begin{aligned} & \left(\frac{\partial v^f}{\partial t}, \varphi^f \right)_{0,f} + \left(\frac{\partial^2 u^s}{\partial t^2}, \varphi^s \right)_{0,s} + ((v^f \cdot \nabla) v^f, \varphi^f)_{0,f} + (\varepsilon(v^f), \varepsilon(\varphi^f))_{0,f} \\ & \quad + (\sigma_s(u^s), \varepsilon(\varphi^s))_{0,s} + \langle \mathbf{M}\bar{v}, \bar{\varphi} \rangle_\Sigma + \langle \mathbf{S}[[v]], [[\varphi]] \rangle_\Sigma \\ & = (\mathbf{f}^f, \varphi^f)_{0,f} + (\mathbf{f}^s, \varphi^s)_{0,s} + \langle \mathbf{g}_N^f, \varphi^f \rangle_{\Gamma_f^N} + \langle \mathbf{h}, \bar{\varphi} \rangle_\Sigma + \langle \mathbf{g}, [[\varphi]] \rangle_\Sigma. \end{aligned} \quad (3.6)$$

Remark 3.2. At the level of weak solutions, the normal traces of the stress tensors $\sigma_s(u^s) \cdot n$ and $\sigma_f(v^f, p) \cdot n$ are, in general, not well-defined as elements of $H^{-1/2}(\Sigma)^d$ based solely on the natural energy regularity of the solution.

More precisely, the existence of the normal trace $\sigma_s(u^s) \cdot n$ for solutions of the elastodynamics equation does not follow from standard trace theorems, but rather from hidden regularity properties of hyperbolic (wave) equation, as established in the works of Lasiecka et al [8]. On the other hand, the normal trace $\sigma_f(v^f, p) \cdot n$ is, in general, not defined at the energy level.

However, such trace regularity is not required for the formulation or analysis of the problem. The weak formulation (3.6) is written in a purely variational form, where the interface terms are defined through duality pairings with test functions. In particular, these terms are expressed via integrals over Σ involving traces of the test functions, which belong to $H^{1/2}(\Sigma)^d$, so that no explicit definition of stress traces is required.

This variational structure ensures that all interface contributions are well-defined under the natural energy regularity and is stable under weak convergence, which is sufficient for passing to the limit in the approximate problem and for establishing the existence of weak solutions.

Let us now introduce the following operators $B : V_f \rightarrow (V_f)'$ and $A_B : V \rightarrow V'$ defined, for $v = (v^f, u^s)$, by:

$$\begin{aligned} \left(\frac{\partial v}{\partial t}, \varphi \right) & \equiv \left(\frac{\partial v^f}{\partial t}, \varphi^f \right)_{0,f} + \left(\frac{\partial v^s}{\partial t}, \varphi^s \right)_{0,s} = \left(\frac{\partial v^f}{\partial t}, \varphi^f \right)_{0,f} + \left(\frac{\partial^2 u^s}{\partial t^2}, \varphi^s \right)_{0,s}, \\ (Bv, \varphi) & \equiv ((v^f \cdot \nabla) v^f, \varphi^f)_{0,f}, \\ (A_B(v), \varphi) & \equiv (\varepsilon(v^f), \varepsilon(\varphi^f))_{0,f} + (\sigma(u^s), \varepsilon(\varphi^s))_{0,s} \\ & \quad + \langle \mathbf{M}\bar{v}, \bar{\varphi} \rangle_\Sigma + \langle \mathbf{S}[[v]], [[\varphi]] \rangle_\Sigma + (Bv, \varphi), \\ (\mathbf{F}, \varphi) & \equiv (\mathbf{f}, \varphi)_0 + \langle \mathbf{g}_N, \varphi^f \rangle_{\Gamma_f^N} + \langle \mathbf{h}, \bar{\varphi} \rangle_\Sigma + \langle \mathbf{g}, [[\varphi]] \rangle_\Sigma. \end{aligned}$$

Replacing the above operators in (3.6) gives the compact form:

$$\left(\frac{\partial v}{\partial t}, \varphi \right) + (A_B(v), \varphi) = (\mathbf{F}, \varphi), \quad (3.7)$$

for all $\varphi = (\varphi^f, \varphi^s) \in L^2(0, T; V)$.

4. Existence of Solution.

4.1. Auxiliary Problem and Abstract Nonlinear Evolution. The proof of existence is based on semigroup theory. Following the same strategy as in [2], one can establish the existence of a solution to the fluid–structure interaction problem by analyzing the following abstract nonlinear evolution equation derived from (3.7):

$$v_t + A_B(v) = \mathbf{F}, \quad (4.1)$$

where the operator domain $\mathcal{D}(A_B)$ is defined as

$$\mathcal{D}(A_B) = \left\{ v = (v^f, u^s) \in V \mid \llbracket \sigma \cdot \mathbf{n} \rrbracket_\Sigma = \mathbf{M}\bar{v}|_\Sigma - \mathbf{h}, \overline{\sigma \cdot \mathbf{n}}|_\Sigma = \mathbf{S}\llbracket v \rrbracket_\Sigma - \mathbf{g} \right\}. \quad (4.2)$$

To facilitate the analysis, we consider an auxiliary problem in which the nonlinear convective operator B is replaced by a general operator L , and accordingly A_B is replaced by A_L . The operator $L : V \rightarrow V'$ is assumed to satisfy the same conditions as in [2]: it is locally Lipschitz, with $L(0) = 0$, and for every $\eta > 0$, there exists a constant $C_\eta > 0$ such that for all $u, v \in V_f$,

$$|(Lu - Lv, u - v)_{0,f}| \leq \eta \|u - v\|_{1,f}^2 + C_\eta \|u - v\|_{0,f}^2. \quad (4.3)$$

The first step is to prove that the operator A_L generates a C_0 -semigroup on V . This is the aim of the following lemma.

Lemma 4.1. *Let L be a nonlinear operator satisfying property (4.3). Then, the operator A_L , with domain $\mathcal{D}(A_L)$ as defined in (4.2), generates a strongly continuous (nonlinear) semigroup on V .*

Proof. The proof is established once we show that A_L is ω -m-accretive.

ω -accretivity: Assume $w = w_1 - w_2 = \left(\underbrace{v_1^f - v_2^f}_{v^f}, \underbrace{u_1^s - u_2^s}_{u^s} \right) \in \mathcal{D}(A_L)$, then:

$$\begin{aligned} (A_L w, w) &= (A_L w_1 - A_L w_2, w) \\ &= (\varepsilon(v^f), \varepsilon(v^f))_{0,f} + (\sigma_s(u^s), \varepsilon(u^s))_{0,s} + \langle \mathbf{M}\bar{v}, \bar{v} \rangle_\Sigma + \langle \mathbf{S}\llbracket v \rrbracket, \llbracket v \rrbracket \rangle_\Sigma \\ &\quad + \left(Lv_1^f - Lv_2^f, v_1^f - v_2^f \right)_{0,f} \\ &\geq \|v^f\|_{1,f}^2 - \eta \|v^f\|_{1,f}^2 - C_\eta \|v^f\|_{0,f}^2. \end{aligned}$$

We now consider the operator $\omega \mathbf{I} + A_L$ instead of A_L . Then we have:

$$\begin{aligned} ((\omega \mathbf{I} + A_L)w, w) &\geq \omega \|v^f\|_{0,f}^2 + \|v^f\|_{1,f}^2 - \eta \|v^f\|_{1,f}^2 - C_\eta \|v^f\|_{0,f}^2 \\ &= (1 - \eta) \|v^f\|_{1,f}^2 + (\omega - C_\eta) \|v^f\|_{0,f}^2. \end{aligned}$$

By choosing $\eta > 0$ and $\omega > 0$ such that $1 - \eta > 0$ and $\omega - C_\eta > 0$, it follows that:

$$(A_L w, w) + \omega \|w\|_0^2 \geq 0, \quad (4.4)$$

which shows that A_L is ω -accretive.

Maximality: The idea is to prove that $\alpha\mathbf{I} + A_L$ is surjective onto $V_f \times V_s$ for α large enough. To that end, we rewrite A_L as:

$$\begin{aligned} (A_L w, \varphi) &= (\varepsilon(v^f), \varepsilon(\varphi^f))_{0,f} + (\sigma_s(u^s), \varepsilon(\varphi^s))_{0,s} + \langle \mathbf{M}\bar{v}, \bar{\varphi} \rangle_\Sigma + \langle \mathbf{S}[[v]], [[\varphi]] \rangle_\Sigma \\ &\quad + (Lv^f, \varphi^f)_{0,f} + \langle \sigma_s \cdot \mathbf{n}_s, \varphi^s \rangle_\Sigma - \langle \sigma_s \cdot \mathbf{n}_s, \varphi^s \rangle_\Sigma \\ &= (\varepsilon(v^f), \varepsilon(\varphi^f))_{0,f} + \langle \sigma_s \cdot \mathbf{n}_s, \varphi^s \rangle_\Sigma + \langle \mathbf{M}\bar{v}, \bar{\varphi} \rangle_\Sigma + \langle \mathbf{S}[[v]], [[\varphi]] \rangle_\Sigma \\ &\quad + (\mathbf{L}v, \varphi)_0 + (\nabla \cdot \sigma_s(u^s), \varphi^s)_{0,s}. \end{aligned}$$

The operator $\alpha\mathbf{I} + A_L$ is surjective if for all $\mathbf{y} \in V_f \times V_s$, there exists $v \in V_f \times V_s$ such that

$$\alpha v + A_L(v) = \mathbf{y}. \quad (4.5)$$

Expanding and writing in weak form for $(\mathbf{y}, \varphi)_0 = (\mathbf{y}^f, \varphi^f)_{0,f} + (\mathbf{y}^s, \varphi^s)_{0,s}$ gives:

$$\begin{aligned} (\alpha v^f, \varphi^f)_{0,f} + (\varepsilon(v^f), \varepsilon(\varphi^f))_{0,f} + \langle \sigma_s \cdot \mathbf{n}_s, \varphi^s \rangle_\Sigma + \langle \mathbf{M}\bar{v}, \bar{\varphi} \rangle_\Sigma + \langle \mathbf{S}[[v]], [[\varphi]] \rangle_\Sigma \\ + (Lv, \varphi)_{0,f} + (\alpha u^s, \varphi^s)_{0,s} + (\nabla \cdot \sigma_s(u^s), \varphi^s)_{0,s} \\ = (\mathbf{y}^f, \varphi^f)_{0,f} + (\mathbf{y}^s, \varphi^s)_{0,s}. \end{aligned} \quad (4.6)$$

The two subproblems derived from (4.6) are:

$$(\alpha u^s, \varphi^s)_{0,s} + (\sigma_s(u^s), \varepsilon(\varphi^s))_{0,s} = (\mathbf{y}^s, \varphi^s)_{0,s}, \quad (4.7)$$

and

$$\begin{aligned} (\alpha v^f, \varphi^f)_{0,f} + (\varepsilon(v^f), \varepsilon(\varphi^f))_{0,f} + \langle \sigma_s \cdot \mathbf{n}_s, \varphi^s \rangle_\Sigma + \langle \mathbf{M}\bar{v}, \bar{\varphi} \rangle_\Sigma + \langle \mathbf{S}[[v]], [[\varphi]] \rangle_\Sigma \\ + (Lv^f, \varphi^f)_{0,f} \\ = (\mathbf{y}^f, \varphi^f)_{0,f}. \end{aligned} \quad (4.8)$$

The weak formulation (4.7) is equivalent to the following classical problem with Fourier-type boundary condition derived from (2.4):

$$\alpha u^s + \nabla \cdot \sigma_s(u^s) = \mathbf{y}^s \quad \text{in } \Omega_s, \quad (4.9)$$

$$\sigma_s \cdot \mathbf{n}_s + \left(\mathbf{S} + \frac{1}{4} \mathbf{M} \right) \frac{\partial u^s}{\partial t} = \mathbf{y}_\Sigma \quad \text{on } \Sigma, \quad (4.10)$$

where

$$\mathbf{y}_\Sigma \equiv \left(\mathbf{S} - \frac{1}{4} \mathbf{M} \right) v^f - \mathbf{g} + \frac{1}{2} \mathbf{h}.$$

Elliptic theory ensures that, given $v^f \in V_f$ and $\mathbf{y}^s \in V_s$ (with $\mathbf{h}$, and $\mathbf{g}$ known functions, and $\mathbf{S}, \mathbf{M}$ satisfy (2.2)), there exists $u^s \in V_s$ solving the problem (4.9)–(4.10).

Since the existence of u^s is guaranteed, we now seek $v^f \in V_f$ satisfying:

$$\begin{aligned} (\alpha v^f, \varphi^f)_{0,f} + (\varepsilon(v^f), \varepsilon(\varphi^f))_{0,f} + \langle \sigma_s \cdot \mathbf{n}_s, \varphi^s \rangle_\Sigma + \langle \mathbf{M}\bar{v}, \bar{\varphi} \rangle_\Sigma + \langle \mathbf{S}[[v]], [[\varphi]] \rangle_\Sigma \\ + (Lv^f, \varphi^f)_{0,f} \\ = (\mathbf{y}^f, \varphi^f)_{0,f}. \end{aligned} \quad (4.11)$$

Note that the quantities $\bar{v}$, $[[v]]$, and $\sigma_s \cdot \mathbf{n}_s$ all involve u^s . However, we cannot move them to the right-hand side, as the solution u^s of the elliptic problem (4.9)–(4.10) depends on v^f .

To prove the existence of v^f solving (4.11), it suffices to show that the operator $R_\alpha^f + L$ is continuous, monotone, and coercive with respect to v^f , where R_α^f is defined as:

$$R_\alpha^f(v^f, \varphi^f) \equiv (\alpha v^f, \varphi^f)_{0,f} + (\varepsilon(v^f), \varepsilon(\varphi^f))_{0,f} + \langle \sigma_s \cdot \mathbf{n}_s, \varphi^s \rangle_\Sigma + \langle \mathbf{M}\bar{v}, \bar{\varphi} \rangle_\Sigma + \langle \mathbf{S}[[v]], [[\varphi]] \rangle_\Sigma. \quad (4.12)$$

Continuity of $R_\alpha^f + L$: To establish continuity, we replace the interface terms in R_α^f by their equivalent expression from the identity (3.5), namely $\langle \sigma_f \cdot \mathbf{n}_f, \varphi^f \rangle_\Sigma$. Thus, the operator $R_\alpha^f(v^f, \varphi^f)$ becomes

$$R_\alpha^f(v^f, \varphi^f) = (\alpha v^f, \varphi^f)_{0,f} + (\varepsilon(v^f), \varepsilon(\varphi^f))_{0,f} + \langle \sigma_f \cdot \mathbf{n}_f, \varphi^f \rangle_\Sigma.$$

We now estimate the expression to prove continuity:

$$\begin{aligned} |R_\alpha^f(v^f, \varphi^f)| &\leq \alpha \|v^f\|_{0,f} \|\varphi^f\|_{0,f} + \|v^f\|_{1,f} \|\varphi^f\|_{1,f} + \|\sigma_f \cdot \mathbf{n}_f\|_{-1/2,\Sigma} \|\varphi^f\|_{1/2,\Sigma} \\ &\leq C_\alpha \|v^f\|_{1,f} \|\varphi^f\|_{1,f}, \end{aligned}$$

where $C_\alpha > 0$ is a constant depending on α . This proves that R_α^f is continuous in V_f . Since $L : V_f \rightarrow V_f'$ is locally Lipschitz, the operator $R_\alpha^f + L$ is continuous on V_f .

Coercivity of $R_\alpha^f + L$: To establish coercivity, it is more convenient to work with the original form of $R(v^f, \varphi^f)$.

$$\begin{aligned} R_\alpha^f(v^f, v^f) + (Lv^f, v^f)_{0,f} &= (\alpha v^f, v^f)_{0,f} + (\varepsilon(v^f), \varepsilon(v^f))_{0,f} + \langle \mathbf{M}\bar{v}, \bar{v} \rangle_\Sigma \\ &\quad + \langle \mathbf{S}[[v]], [[v]] \rangle_\Sigma + \langle \sigma_s \cdot \mathbf{n}_s, u^s \rangle_\Sigma + (Lv^f, v^f)_{0,f}. \end{aligned} \quad (4.13)$$

We first estimate the term $\langle \sigma_s \cdot \mathbf{n}_s, u^s \rangle_\Sigma$. To do so, we take $\mathbf{y}^s = 0$ in equation (4.9) and write it in weak form with $\varphi^s = u^s$, which gives:

$$\alpha \|u^s\|_{0,s}^2 + (\sigma_s(u^s), \varepsilon(u^s))_{0,s} - \langle \sigma_s \cdot \mathbf{n}_s, u^s \rangle_\Sigma = 0,$$

that gives the inequality

$$\langle \sigma_s \cdot \mathbf{n}_s, u^s \rangle_\Sigma = \alpha \|u^s\|_{0,s}^2 + (\sigma_s(u^s), \varepsilon(u^s))_{0,s} \geq 0. \quad (4.14)$$

Using the bound from (4.3) and inequality (4.14) in (4.13) yields:

$$R(v^f, v^f) \geq (\alpha - C_\eta) \|v^f\|_{0,f}^2 + (1 - \eta) \|v^f\|_{1,f}^2.$$

Thus, for α sufficiently large and η sufficiently small, coercivity of $R_\alpha^f + L$ is ensured. As a result, A_L is ω -m-accretive and hence generates a nonlinear contraction semigroup on $V_f \times V_s$. $\square$

Corollary 4.2. *For every data $(v_0^f, u_0^s, v_0^s) \in \mathcal{D}(A_L) \times V_s$, the semigroup solution (v^f, u^s) satisfies, for all $t \in (0, T)$, the variational formulation (3.7) with L instead of B .*

The proof is straightforward.

Theorem 4.3. *Let $(v_0^f, u_0^s, v_0^s) \in V_f \times V_s \times V_s$, and let (v^f, u^s) be the corresponding semigroup solutions given by Lemma 4.1. Then the solutions (v^f, u^s) satisfy the variational formulation (3.7).*

The key point of this theorem is to show that even when the initial data lie in the larger space $V_f \times V_s \times V_s$ (instead of $\mathcal{D}(A_L) \times V_s$), the semigroup solution still satisfies the variational formulation.

Proof. The proof is complete once the density of $\mathcal{D}(A_L)$ in $V_f \times V_s$ is established. Following the same strategy as in [2], consider the operator $\mathbf{I} + \frac{1}{\alpha}A_L$, which is surjective for a given L (by Lemma 4.1). Given any $\mathbf{y} \in V_f \times V_s$, there exists $\mathbf{U}_{L,\alpha} = (v^f, u^s) \in \mathcal{D}(A_L)$ such that

$$\mathbf{U}_{L,\alpha} + \frac{1}{\alpha}A_L\mathbf{U}_{L,\alpha} = \mathbf{y}.$$

Taking the limit $\alpha \rightarrow \infty$ yields

$$\lim_{\alpha \rightarrow \infty} \mathbf{U}_{L,\alpha} = \mathbf{y},$$

which proves that $\mathcal{D}(A_L)$ is dense in $V_f \times V_s$. That is, any initial data $\mathbf{U}_0 = (v_0^f, u_0^s, v_0^s) \in V \times V_s$ can be approximated by a sequence

$$\mathbf{U}_0^n = (v_0^{f,n}, u_0^{s,n}, v_0^s) \in \mathcal{D}(A_L) \times V_s,$$

such that

$$(v_0^{f,n}, u_0^{s,n}) \rightarrow (v_0^f, u_0^s) \quad \text{in } \mathcal{D}(A_L).$$

□

Now, we return to the original operator B but with a more general form:

$$\begin{aligned} B(u, v) &\equiv (u \cdot \nabla)v, \\ B(u) &\equiv B(u, u), \\ b(u, v, w) &\equiv ((u \cdot \nabla)v, w) \end{aligned} \tag{4.15}$$

Let us also consider a truncation of the nonlinear operator B , and let it be $B_n : V_f \rightarrow V_f'$ defined by

$$B_n(v) \equiv \begin{cases} B(v), & \|v\|_1 \leq n, \\ n^2 \frac{B(v)}{\|v\|_1^2}, & \|v\|_1 \geq n. \end{cases} \tag{4.16}$$

The nonlinear operator B_n verifies property (4.3) and the same properties as L (see Lemma 3.5 of [2] for a detailed proof), meaning that the existence of a semigroup solution corresponding to the operator $A_n \equiv A_{B_n}$ is guaranteed. It satisfies the following weak formulation:

$$\begin{aligned} \left(\frac{\partial v^{f,n}}{\partial t}, \varphi^f \right)_{0,f} + \left(\frac{\partial^2 u^{s,n}}{\partial t^2}, \varphi^s \right)_{0,s} + (\varepsilon(v^{f,n}), \varepsilon(\varphi^f))_{0,f} + (\sigma_s(u^{s,n}), \varepsilon(\varphi^s))_{0,s} \\ + (B_n(v^{f,n}), \varphi^f)_{0,f} + \langle \mathbf{M} \overline{v^n}, \overline{\varphi} \rangle_\Sigma + \langle \mathbf{S} \llbracket v^n \rrbracket, \llbracket \varphi \rrbracket \rangle_\Sigma \\ = (\mathbf{f}^f, \varphi^f)_{0,f} + (\mathbf{f}^s, \varphi^s)_{0,s} + \langle \mathbf{g}_N^f, \varphi^f \rangle_{\Gamma_f^N} + \langle \mathbf{h}, \overline{\varphi} \rangle_\Sigma + \langle \mathbf{g}, \llbracket \varphi \rrbracket \rangle_\Sigma \end{aligned} \tag{4.17}$$

What remains is to prove that, by passing to the limit as $n \rightarrow \infty$, the result remains valid. This is the goal of the next theorem.

Before that, we state the following lemma from [2] regarding the properties of the trilinear form $b(\cdot, \cdot, \cdot)$.

Lemma 4.4 ([2, 11]). *The trilinear form $b(\cdot, \cdot, \cdot)$, given in (4.15), satisfies the following properties for all $u \in V_f$, $v, w \in [H^1(\Omega_f)]^d$, where $d = \dim(\Omega_f)$:*

1. For $p_1, p_2, p_3 \geq 0$ such that

$$\begin{cases} p_1 + p_2 + p_3 \geq \frac{d}{2}, & \text{if } p_i \neq \frac{d}{2} \text{ for all } i \in \{1, 2, 3\}, \\ p_1 + p_2 + p_3 > \frac{d}{2}, & \text{if } p_i = \frac{d}{2} \text{ for some } i \in \{1, 2, 3\}, \end{cases}$$

then there exists a constant $C > 0$ such that

$$|((u \cdot \nabla)v, w)| \leq C \|u\|_{p_1, f} \|v\|_{p_2+1, f} \|w\|_{p_3, f}.$$

2. If $d = 2$, then there exist constants $C_1, C_2 > 0$ such that

$$|b(u, v, w)| \leq C_1 \|u\|_{1/2, f} \|v\|_{1, f} \|w\|_{1/2, f} + C_2 \|u\|_{1/2, f} \|v\|_{3/4, f} \|w\|_{3/4, f}.$$

3. If $d = 3$, then there exist constants $C_3, C_4 > 0$ such that

$$\begin{aligned} |b(u, v, w)| &\leq C_3 \|u\|_{1/2, f} \|v\|_{1, f} \|w\|_{1, f}, \\ |b(u, v, w)| &\leq C_4 \|u\|_{3/4, f} \|v\|_{1, f} \|w\|_{3/4, f}. \end{aligned}$$

4. The trilinear form satisfies the skew-symmetry property:

$$b(u, v, w) = -b(u, w, v), \quad \text{in particular, } b(u, u, u) = 0.$$

For the proof of Lemma 4.4, see [2, 11].

We also need to present the following result, which will be needed in the proof of the next theorem.

Proposition 4.5. Let $u^s, v^s = \frac{\partial u^s}{\partial t} \in \mathbb{H}^1(\Omega_s)$. Let $\sigma_s(u^s)$ and $\varepsilon(u^s)$ be the linear elasticity stress tensor and the symmetric strain tensor, respectively. Then:

$$(\sigma_s(u^s), \varepsilon(v^s))_{0, s} = (\varepsilon(v^s), \sigma_s(u^s))_{0, s} \quad \text{and} \quad (\sigma_s(u^s), \varepsilon(v^s))_{0, s} = (\varepsilon(u^s), \sigma_s(v^s))_{0, s}.$$

and

$$\begin{aligned} \frac{1}{2} \frac{d}{dt} (\sigma_s(u^{s, n}(s)), \varepsilon(u^{s, n}(s)))_{0, s} &= \frac{1}{2} (\sigma_s(v^{s, n}(s)), \varepsilon(u^{s, n}(s)))_{0, s} \\ &\quad + \frac{1}{2} (\sigma_s(u^{s, n}(s)), \varepsilon(v^{s, n}(s)))_{0, s} \\ &= \frac{1}{2} (\varepsilon(v^{s, n}(s)), \sigma_s(u^{s, n}(s)))_{0, s} \\ &\quad + \frac{1}{2} (\sigma_s(u^{s, n}(s)), \varepsilon(v^{s, n}(s)))_{0, s} \\ &= (\sigma_s(u^{s, n}(s)), \varepsilon(v^{s, n}(s)))_{0, s}. \end{aligned}$$

Theorem 4.6. For every initial condition $(\mathbf{v}_0^f, \mathbf{v}_0^s, \mathbf{u}_0^s) \in V_f \times V_s \times V_s$, there exists a solution

$$(\mathbf{v}^f, \mathbf{v}^s) \in C([0, T]; H_f \times H_s) \cap L^2(0, T; V_f \times V_s),$$

with

$$\partial_t \mathbf{v}^s \in L^1(0, T; V_s'), \quad \partial_t \mathbf{v}^f \in L^1(0, T; V_f'), \quad B(\mathbf{v}^f) \in L^1(0, T; V_f'),$$

and

$$\mathbf{u}^s \in H^1(0, T; V_s), \quad \partial_t \mathbf{u}^s = \mathbf{v}^s,$$

such that for all $(\varphi^f, \varphi^s) \in V_f \times V_s$ and for a.e. $t \in (0, T)$,

$$\begin{aligned} & \left\langle \frac{\partial \mathbf{v}^f}{\partial t}, \varphi^f \right\rangle_{V_f', V_f} + \left\langle \frac{\partial \mathbf{v}^s}{\partial t}, \varphi^s \right\rangle_{V_s', V_s} + (\varepsilon(\mathbf{v}^f), \varepsilon(\varphi^f))_{0,f} \\ & \quad + (\sigma_s(\mathbf{u}^s), \varepsilon(\varphi^s))_{0,s} + \langle B(\mathbf{v}^f), \varphi^f \rangle_{V_f', V_f} \\ & \quad + \langle M\bar{\mathbf{v}}, \bar{\varphi} \rangle_\Sigma + \langle S[\mathbf{v}], [\varphi] \rangle_\Sigma \\ & = (\mathbf{f}^f, \varphi^f)_{0,f} + (\mathbf{f}^s, \varphi^s)_{0,s} + \langle \mathbf{g}_N^f, \varphi^f \rangle_{\Gamma_f^N} + \langle \mathbf{h}, \bar{\varphi} \rangle_\Sigma + \langle \mathbf{g}, [\varphi] \rangle_\Sigma. \end{aligned} \quad (4.18)$$

Proof. From Theorem 4.3, one can conclude the existence of semigroup solutions $(\mathbf{v}^{s,n}, \mathbf{v}^{f,n}) \in V_s \times V_f$. The desired result should be achieved once we take $n \rightarrow \infty$ in the weak formulation (4.17). For that, some a priori bounds are needed.

Step 1: A priori bounds. Let us consider the weak form (4.17) with the test functions $\varphi^f = \mathbf{v}^{f,n}$, $\varphi^s = \mathbf{v}^{s,n}$. Using the property $b(\mathbf{v}^{f,n}, \mathbf{v}^{f,n}, \mathbf{v}^{f,n}) = 0$, we obtain:

$$\begin{aligned} & \frac{1}{2} \frac{d}{dt} \left(\|\mathbf{v}^{f,n}(t)\|_{0,f}^2 + \|\mathbf{v}^{s,n}(t)\|_{0,s}^2 \right) + (\sigma_s(\mathbf{u}^{s,n}(t)), \varepsilon(\mathbf{v}^{s,n}(t)))_{0,s} + \|\varepsilon(\mathbf{v}^{f,n}(t))\|_{0,f}^2 \\ & \quad + \langle M\bar{\mathbf{v}}^n(t), \bar{\mathbf{v}}^n(t) \rangle_\Sigma + \langle S[\mathbf{v}^n(t)], [\mathbf{v}^n(t)] \rangle_\Sigma \\ & = (\mathbf{f}^f(t), \mathbf{v}^{f,n}(t))_{0,f} + (\mathbf{f}^s(t), \mathbf{v}^{s,n}(t))_{0,s} + \langle \mathbf{g}_N^f(t), \mathbf{v}^{f,n}(t) \rangle_{\Gamma_f^N} \\ & \quad + \langle \mathbf{h}(t), \bar{\mathbf{v}}^n(t) \rangle_\Sigma + \langle \mathbf{g}(t), [\mathbf{v}^n(t)] \rangle_\Sigma. \end{aligned} \quad (4.19)$$

Applying the ellipticity assumption (2.2) on M and S , and integrating over the time interval $(0, t)$, for almost every $t \in (0, T)$, yields:

$$\begin{aligned} & \frac{1}{2} \left(\|\mathbf{v}^{f,n}(t)\|_{0,f}^2 + \|\mathbf{v}^{s,n}(t)\|_{0,s}^2 \right) + \int_0^t \|\varepsilon(\mathbf{v}^{f,n}(s))\|_{0,f}^2 ds + M_0 \int_0^t \|\bar{\mathbf{v}}^n(s)\|_{1/2,\Sigma}^2 ds \\ & \quad + \int_0^t (\sigma_s(\mathbf{u}^{s,n}(s)), \varepsilon(\mathbf{v}^{s,n}(s)))_{0,s} ds + S_0 \int_0^t \|[\mathbf{v}^n(s)]\|_{1/2,\Sigma}^2 ds \\ & \leq \int_0^t (\mathbf{f}^f(s), \mathbf{v}^{f,n}(s))_{0,f} ds + \int_0^t (\mathbf{f}^s(s), \mathbf{v}^{s,n}(s))_{0,s} ds + \int_0^t \langle \mathbf{g}_N^f(s), \mathbf{v}^{f,n}(s) \rangle_{\Gamma_f^N} ds \\ & \quad + \int_0^t \langle \mathbf{h}(s), \bar{\mathbf{v}}^n(s) \rangle_\Sigma ds + \int_0^t \langle \mathbf{g}(s), [\mathbf{v}^n(s)] \rangle_\Sigma ds + \frac{1}{2} \left(\|\mathbf{v}_0^f\|_{0,f}^2 + \|\mathbf{v}_0^s\|_{0,s}^2 \right). \end{aligned} \quad (4.20)$$

Using *ii.* from proposition 4.5 in inequality (4.20) we get

$$\begin{aligned} & \frac{1}{2} \left(\|\mathbf{v}^{f,n}(t)\|_{0,f}^2 + \|\mathbf{v}^{s,n}(t)\|_{0,s}^2 \right) + \int_0^t \|\varepsilon(\mathbf{v}^{f,n}(s))\|_{0,f}^2 ds + M_0 \int_0^t \|\bar{\mathbf{v}}^n(s)\|_{1/2,\Sigma}^2 ds \\ & \quad + \frac{1}{2} (\sigma_s(\mathbf{u}^{s,n}(t)), \varepsilon(\mathbf{u}^{s,n}(t)))_{0,s} + S_0 \int_0^t \|[\mathbf{v}^n(s)]\|_{1/2,\Sigma}^2 ds \\ & \leq \int_0^t (\mathbf{f}^f(s), \mathbf{v}^{f,n}(s))_{0,f} ds + \int_0^t (\mathbf{f}^s(s), \mathbf{v}^{s,n}(s))_{0,s} ds + \int_0^t \langle \mathbf{g}_N^f(s), \mathbf{v}^{f,n}(s) \rangle_{\Gamma_f^N} ds \\ & \quad + \int_0^t \langle \mathbf{h}(s), \bar{\mathbf{v}}^n(s) \rangle_\Sigma ds + \int_0^t \langle \mathbf{g}(s), [\mathbf{v}^n(s)] \rangle_\Sigma ds \\ & \quad + \frac{1}{2} \left(\|\mathbf{v}_0^f\|_{0,f}^2 + \|\mathbf{v}_0^s\|_{0,s}^2 + (\sigma_s(\mathbf{u}_0^s), \varepsilon(\mathbf{u}_0^s))_{0,s} \right). \end{aligned} \quad (4.21)$$

Applying Cauchy-Schwarz inequality to the right-hand side terms in (4.21) gives:

$$\begin{aligned}
& \frac{1}{2} \left(\|\mathbf{v}^{f,n}(t)\|_{0,f}^2 + \|\mathbf{v}^{s,n}(t)\|_{0,s}^2 \right) + \int_0^t \|\varepsilon(\mathbf{v}^{f,n}(s))\|_{0,f}^2 ds + \mathbf{M}_0 \int_0^t \|\overline{\mathbf{v}^n(s)}\|_{1/2,\Sigma}^2 ds \\
& \quad + \frac{1}{2} (\sigma_s(\mathbf{u}^{s,n}(t)), \varepsilon(\mathbf{u}^{s,n}(t)))_{0,s} + \mathbf{S}_0 \int_0^t \|\llbracket \mathbf{v}^n(s) \rrbracket\|_{1/2,\Sigma}^2 ds \\
& \leq \int_0^t \|\mathbf{f}^f(s)\|_{0,f} \|\mathbf{v}^{f,n}(s)\|_{0,f} ds + \int_0^t \|\mathbf{f}^s(s)\|_{0,s} \|\mathbf{v}^{s,n}(s)\|_{0,s} ds \\
& \quad + \int_0^t \|\mathbf{h}(s)\|_{1/2,\Sigma} \|\overline{\mathbf{v}^n(s)}\|_{1/2,\Sigma} ds + \int_0^t \|\mathbf{g}(s)\|_{1/2,\Sigma} \|\llbracket \mathbf{v}^n(s) \rrbracket\|_{1/2,\Sigma} ds \\
& \quad + \int_0^t \|\mathbf{g}_N^f(s)\|_{1/2,\Gamma_f^N} \|\mathbf{v}^{f,n}(s)\|_{1/2,\Gamma_f^N} ds \\
& \quad + \frac{1}{2} \left(\|\mathbf{v}_0^f\|_{0,f}^2 + \|\mathbf{v}_0^s\|_{0,s}^2 + (\sigma_s(\mathbf{u}_0^s), \varepsilon(\mathbf{u}_0^s))_{0,s} \right)
\end{aligned} \tag{4.22}$$

From the coercivity of the elasticity tensor together with Korn's inequality (see [4]), we know that

$$\begin{aligned}
& (\sigma_s(\mathbf{u}^{s,n}(t)), \varepsilon(\mathbf{u}^{s,n}(t)))_{0,s} \geq c_s \|\mathbf{u}^{s,n}(t)\|_{1,s}^2 \\
& \quad \text{and} \\
& \|\varepsilon(\mathbf{v}^{f,n}(t))\|_{0,f}^2 \geq c_6 \|\mathbf{v}^{f,n}(t)\|_{1,f}^2.
\end{aligned} \tag{4.23}$$

Applying (4.23) to inequality (4.22) yields

$$\begin{aligned}
& \frac{1}{2} \left(\|\mathbf{v}^{f,n}(t)\|_{0,f}^2 + \|\mathbf{v}^{s,n}(t)\|_{0,s}^2 \right) + c_6 \int_0^t \|\mathbf{v}^{f,n}(s)\|_{1,f}^2 ds \\
& \quad + \frac{1}{4} (\sigma_s(\mathbf{u}^{s,n}(t)), \varepsilon(\mathbf{u}^{s,n}(t)))_{0,s} + \frac{c_5}{4} \|\mathbf{u}^{s,n}(t)\|_{1,s}^2 \\
& \quad + \mathbf{M}_0 \int_0^t \|\overline{\mathbf{v}^n(s)}\|_{1/2,\Sigma}^2 ds + \mathbf{S}_0 \int_0^t \|\llbracket \mathbf{v}^n(s) \rrbracket\|_{1/2,\Sigma}^2 ds \\
& \leq \int_0^t \|\mathbf{f}^f(s)\|_{0,f} \|\mathbf{v}^{f,n}(s)\|_{0,f} ds + \int_0^t \|\mathbf{f}^s(s)\|_{0,s} \|\mathbf{v}^{s,n}(s)\|_{0,s} ds \\
& \quad + \int_0^t \|\mathbf{h}(s)\|_{1/2,\Sigma} \|\overline{\mathbf{v}^n(s)}\|_{1/2,\Sigma} ds + \int_0^t \|\mathbf{g}(s)\|_{1/2,\Sigma} \|\llbracket \mathbf{v}^n(s) \rrbracket\|_{1/2,\Sigma} ds \\
& \quad + \int_0^t \|\mathbf{g}_N^f(s)\|_{1/2,\Gamma_f^N} \|\mathbf{v}^{f,n}(s)\|_{1/2,\Gamma_f^N} ds \\
& \quad + \frac{1}{2} \left(\|\mathbf{v}_0^f\|_{0,f}^2 + \|\mathbf{v}_0^s\|_{0,s}^2 + (\sigma_s(\mathbf{u}_0^s), \varepsilon(\mathbf{u}_0^s))_{0,s} \right)
\end{aligned} \tag{4.24}$$

By applying Young's inequality $ab \leq \frac{1}{2c}a^2 + \frac{c}{2}b^2$, we obtain:

$$\begin{aligned}
& \frac{1}{2} \left(\|\mathbf{v}^{f,n}(t)\|_{0,f}^2 + \|\mathbf{v}^{s,n}(t)\|_{0,s}^2 \right) + c_6 \int_0^t \|\mathbf{v}^{f,n}(s)\|_{1,f}^2 ds \\
& \quad + \frac{1}{4} (\sigma_s(\mathbf{u}^{s,n}(t)), \varepsilon(\mathbf{u}^{s,n}(t)))_{0,s} + \frac{c_5}{4} \|\mathbf{u}^{s,n}(t)\|_{1,s}^2
\end{aligned}$$

$$\begin{aligned}
& + M_0 \int_0^t \|\overline{\mathbf{v}^n(s)}\|_{1/2,\Sigma}^2 ds + S_0 \int_0^t \|\llbracket \mathbf{v}^n(s) \rrbracket\|_{1/2,\Sigma}^2 ds \\
& \leq \frac{1}{2c_7} \int_0^t \|\mathbf{f}^f(s)\|_{0,f}^2 ds + \frac{c_7}{2} \int_0^t \|\mathbf{v}^{f,n}(s)\|_{0,f}^2 ds \\
& + \frac{1}{2c_8} \int_0^t \|\mathbf{f}^s(s)\|_{0,s}^2 ds + \frac{c_8}{2} \int_0^t \|\mathbf{v}^{s,n}(s)\|_{0,s}^2 ds \\
& + \frac{1}{2c_9} \int_0^t \|\mathbf{h}(s)\|_{1/2,\Sigma}^2 ds + \frac{c_9}{2} \int_0^t \|\overline{\mathbf{v}^n(s)}\|_{1/2,\Sigma}^2 ds \\
& + \frac{1}{2c_{10}} \int_0^t \|\mathbf{g}(s)\|_{1/2,\Sigma}^2 ds + \frac{c_{10}}{2} \int_0^t \|\llbracket \mathbf{v}^n(s) \rrbracket\|_{1/2,\Sigma}^2 ds \\
& + \frac{1}{2c_{11}} \int_0^t \|\mathbf{g}_N^f(s)\|_{1/2,\Gamma_f^N}^2 ds + \frac{c_{11}}{2} \int_0^t \|\mathbf{v}^{f,n}(s)\|_{1/2,\Gamma_f^N}^2 ds \\
& + \frac{1}{2} \left(\|\mathbf{v}_0^f\|_{0,f}^2 + \|\mathbf{v}_0^s\|_{0,s}^2 + (\sigma_s(\mathbf{u}_0^s), \varepsilon(\mathbf{u}_0^s))_{0,s} \right) \tag{4.25}
\end{aligned}$$

Using the following trace and a Poincaré-type inequalities:

$$\|\mathbf{v}^{f,n}\|_{1/2,\Gamma_f^N} \leq c_{12} \|\mathbf{v}^{f,n}\|_{1,f} \text{ and } \|\mathbf{v}^{f,n}\|_{0,f} \leq c_{13} \|\mathbf{v}^{f,n}\|_{1,f}, \tag{4.26}$$

we get after summing the known terms in one term denoted by $F = F(f^f, f^s, g_N^f, h, g)$ and for a constant $C = C(c_7, c_8, c_9, c_{10}, c_{11}) > 0$

$$\begin{aligned}
& \frac{1}{2} \left(\|\mathbf{v}^{f,n}(t)\|_{0,f}^2 + \|\mathbf{v}^{s,n}(t)\|_{0,s}^2 \right) + c_6 \int_0^t \|\mathbf{v}^{f,n}(s)\|_{1,f}^2 ds + \frac{1}{4} (\sigma_s(\mathbf{u}^{s,n}(t)), \varepsilon(\mathbf{u}^{s,n}(t)))_{0,s} \\
& + \frac{c_5}{4} \|\mathbf{u}^{s,n}(t)\|_{1,s}^2 + M_0 \int_0^t \|\overline{\mathbf{v}^n(s)}\|_{1/2,\Sigma}^2 ds + S_0 \int_0^t \|\llbracket \mathbf{v}^n(s) \rrbracket\|_{1/2,\Sigma}^2 ds \\
& \leq \left(\frac{c_7 c_{13} + c_{11} c_{12}}{2} \right) \int_0^t \|\mathbf{v}^{f,n}(s)\|_{1,f}^2 ds + \frac{c_8}{2} \int_0^t \|\mathbf{v}^{s,n}(s)\|_{0,s}^2 ds \\
& + \frac{c_9}{2} \int_0^t \|\overline{\mathbf{v}^n(s)}\|_{1/2,\Sigma}^2 ds + \frac{c_{10}}{2} \int_0^t \|\llbracket \mathbf{v}^n(s) \rrbracket\|_{1/2,\Sigma}^2 ds \\
& + \frac{1}{2} \left(\|\mathbf{v}_0^f\|_{0,f}^2 + \|\mathbf{v}_0^s\|_{0,s}^2 + (\sigma_s(\mathbf{u}_0^s), \varepsilon(\mathbf{u}_0^s))_{0,s} \right) + C \int_0^t \mathbf{F}(f^f, f^s, g_N^f, h, g)(s) ds \tag{4.27}
\end{aligned}$$

We now absorb terms from the right-hand side into the left-hand side by comparing the coefficients of the common integrals.

1. Absorbing the fluid dissipation term. We absorb the term

$$\left(\frac{c_7 c_{13} + c_{11} c_{12}}{2} \right) \int_0^t \|\mathbf{v}^{f,n}(s)\|_{1,f}^2 ds$$

into the left-hand side term

$$c_6 \int_0^t \|\mathbf{v}^{f,n}(s)\|_{1,f}^2 ds$$

which is possible if we choose the constants c_7 and c_{11} from Young's inequality to satisfy the inequality

$$c_6 > \frac{c_7 c_{13} + c_{11} c_{12}}{2}.$$

2. Absorbing the interface mean and jump terms. We absorb

$$\frac{c_9}{2} \int_0^t \|\overline{\mathbf{v}^n(s)}\|_{1/2,\Sigma}^2 ds \quad \text{and} \quad \frac{c_{10}}{2} \int_0^t \|[\![\mathbf{v}^n(s)]\!]\|_{1/2,\Sigma}^2 ds,$$

repectively into

$$\mathbf{M}_0 \int_0^t \|\overline{\mathbf{v}^n(s)}\|_{1/2,\Sigma}^2 ds \quad \text{and} \quad \mathbf{S}_0 \int_0^t \|[\![\mathbf{v}^n(s)]\!]\|_{1/2,\Sigma}^2 ds,$$

provided that c_9 and c_{10} are chosen to verify

$$\mathbf{M}_0 > \frac{c_9}{2} \quad \text{and} \quad \mathbf{S}_0 > \frac{c_{10}}{2}.$$

3. Remaining terms. The remaining term

$$\frac{c_8}{2} \int_0^t \|\mathbf{v}^{s,n}(s)\|_{0,s}^2 ds$$

cannot be absorbed and will be handled later using Grönwall's inequality.

We obtain the following absorbed inequality:

$$\begin{aligned} & \frac{1}{2} (\|\mathbf{v}^{f,n}(t)\|_{0,f}^2 + \|\mathbf{v}^{s,n}(t)\|_{0,s}^2) + \frac{c_5}{4} \|\mathbf{u}^{s,n}(t)\|_{1,s}^2 + \frac{1}{4} (\sigma_s(\mathbf{u}^{s,n}(t)), \varepsilon(\mathbf{u}^{s,n}(t)))_{0,s} \\ & + \int_0^t \|\mathbf{v}^{s,n}(s)\|_{1,s}^2 ds + \theta_1 \int_0^t \|\mathbf{v}^{f,n}(s)\|_{1,f}^2 ds + \theta_2 \int_0^t \|\overline{\mathbf{v}^n(s)}\|_{1/2,\Sigma}^2 ds \\ & + \theta_3 \int_0^t \|[\![\mathbf{v}^n(s)]\!]\|_{1/2,\Sigma}^2 ds \\ & \leq \frac{1}{2} (\|\mathbf{v}_0^f\|_{0,f}^2 + \|\mathbf{v}_0^s\|_{0,s}^2 + (\sigma_s(\mathbf{u}_0^s), \varepsilon(\mathbf{u}_0^s))_{0,s}) + \frac{c_8}{2} \int_0^t \|\mathbf{v}^{s,n}(s)\|_{0,s}^2 ds \\ & + C \int_0^t \mathbf{F}(\mathbf{f}^f, \mathbf{f}^s, \mathbf{g}_N^f, \mathbf{h}, \mathbf{g})(s) ds, \end{aligned} \tag{4.28}$$

where:

$$\theta_1 := c_6 - \frac{c_7 c_{13} + c_{11} c_{12}}{2}, \quad \theta_2 := \mathbf{M}_0 - \frac{c_9}{2}, \quad \theta_3 := \mathbf{S}_0 - \frac{c_{10}}{2}.$$

We define the energy functional for every $n > 0$:

$$\begin{aligned} E^n(t) &:= \|\mathbf{v}^{f,n}(t)\|_{0,f}^2 + \|\mathbf{v}^{s,n}(t)\|_{0,s}^2 + \|\mathbf{u}^{s,n}(t)\|_{1,s}^2 + (\sigma_s(\mathbf{u}^{s,n}(t)), \varepsilon(\mathbf{u}^{s,n}(t)))_{0,s} \\ &+ \int_0^t \left[\|\mathbf{v}^{f,n}(s)\|_{1,f}^2 + \|\overline{\mathbf{v}^n(s)}\|_{1/2,\Sigma}^2 + \|[\![\mathbf{v}^n(s)]\!]\|_{1/2,\Sigma}^2 \right] ds. \end{aligned} \tag{4.29}$$

This directly yields the inequality

$$\|\mathbf{v}^{s,n}(t)\|_{0,s}^2 \leq E^n(t). \tag{4.30}$$

which gives

$$\int_0^t \|\mathbf{v}^{s,n}(s)\|_{0,s}^2 ds \leq \int_0^t E^n(s) ds. \tag{4.31}$$

Using inequality (4.28) and (4.30), and noting that all terms on the left-hand side are non-negative, we obtain:

$$E^n(t) \leq C_0 + \frac{c_8}{2} \int_0^t E^n(s) ds + \int_0^t F(s) ds,$$

where:

$$\begin{aligned} C_0 &:= \|\mathbf{v}_0^f\|_{0,f}^2 + \|\mathbf{v}_0^s\|_{0,s}^2 + (\sigma_s(\mathbf{u}_0^s), \varepsilon(\mathbf{u}_0^s))_{0,s}, \\ F(s) &:= C \cdot \mathbf{F}(\mathbf{f}^f, \mathbf{f}^s, \mathbf{g}_N^f, \mathbf{h}, \mathbf{g})(s). \end{aligned}$$

We now apply Grönwall's inequality to get:

$$E^n(t) \leq \left(C_0 + \int_0^t F(s) ds \right) \exp\left(\frac{c_8}{2}t\right).$$

Therefore, the velocities and displacement satisfy the a priori bound:

$$\begin{aligned} &\|\mathbf{v}^{f,n}(t)\|_{0,f}^2 + \|\mathbf{v}^{s,n}(t)\|_{0,s}^2 + \|\mathbf{u}^{s,n}(t)\|_{1,s}^2 + (\sigma_s(\mathbf{u}^{s,n}(t)), \varepsilon(\mathbf{u}^{s,n}(t)))_{0,s} \\ &+ \int_0^t \left[\|\mathbf{v}^{f,n}(s)\|_{1,f}^2 + \|\overline{\mathbf{v}^n(s)}\|_{1/2,\Sigma}^2 + \|\llbracket \mathbf{v}^n(s) \rrbracket\|_{1/2,\Sigma}^2 \right] ds. \\ &\leq \left(\|\mathbf{v}_0^f\|_{0,f}^2 + \|\mathbf{v}_0^s\|_{0,s}^2 + (\sigma_s(\mathbf{u}_0^s), \varepsilon(\mathbf{u}_0^s))_{0,s} + \int_0^t F(s) ds \right) e^{\frac{c_8}{2}T}. \end{aligned} \quad (4.32)$$

Let us now use the notation

$$M(t) := \int_0^t \mathbf{F}(s) ds, \quad \text{and} \quad C_0 := \|\mathbf{v}_0^f\|_{0,f}^2 + \|\mathbf{v}_0^s\|_{0,s}^2 + (\sigma_s(\mathbf{u}_0^s), \varepsilon(\mathbf{u}_0^s))_{0,s} + M(T).$$

We can deduce from inequality (4.32) some a priori bounds:

A priori bounds for $\mathbf{v}^{f,n}$:

$$\|\mathbf{v}^{f,n}(t)\|_{0,f}^2 \leq C_0 e^{\frac{c_8}{2}T}, \quad (4.33)$$

$$\int_0^T \|\mathbf{v}^{f,n}(t)\|_{0,f}^2 ds \leq C_0 T e^{\frac{c_8}{2}T}. \quad (4.34)$$

$$\int_0^t \|\mathbf{v}^{f,n}(t)\|_{1,f}^2 ds \leq C_0 e^{\frac{c_8}{2}T}. \quad (4.35)$$

A priori bounds for $\mathbf{u}^{s,n}$ and $\mathbf{v}^{s,n}$:

$$\|\mathbf{u}^{s,n}(t)\|_{1,s}^2 \leq C_0 e^{\frac{c_8}{2}T}. \quad (4.36)$$

$$\|\mathbf{v}^{s,n}(t)\|_{0,s}^2 \leq C_0 e^{\frac{c_8}{2}T}. \quad (4.37)$$

$$\int_0^T \|\mathbf{v}^{s,n}(t)\|_{0,s}^2 ds \leq C_0 T e^{\frac{c_8}{2}T}. \quad (4.38)$$

A priori bound on the elastic potential energy:

$$(\sigma_s(\mathbf{u}^{s,n}(t)), \varepsilon(\mathbf{u}^{s,n}(t)))_{0,s} \leq C_0 T e^{\frac{c_8}{2}T}. \quad (4.39)$$

The a priori estimates derived above yield the following regularity properties:

$$\begin{aligned} \mathbf{v}^{f,n} &\in L^\infty(0, T; L^2(\Omega_f)) \cap L^2(0, T; V^f), \\ \mathbf{v}^{s,n} &\in L^\infty(0, T; L^2(\Omega_s)), \\ \mathbf{u}^{s,n} &\in L^\infty(0, T; V^s), \\ \overline{\mathbf{v}^n}, \llbracket \mathbf{v}^n \rrbracket &\in L^2(0, T; H^{1/2}(\Sigma)), \end{aligned} \quad (4.40)$$

Step 2: Passing to the limit in the linear terms. From (4.37–4.38), $\mathbf{v}^{s,n}$ is uniformly bounded in $L^2(0, T; V_s)$. Hence, we can extract a subsequence (still denoted $\mathbf{v}^{s,n}$) such that

$$\mathbf{v}^{s,n} \rightharpoonup \mathbf{v}^s \quad \text{in } L^2(0, T; V_s).$$

To justify strong convergence results that are needed later, we invoke classical compactness results. First, the Rellich–Kondrachov theorem ensures that the embedding

$$H^1(\Omega) \hookrightarrow L^2(\Omega)$$

is compact for bounded Lipschitz domains. This implies that sequences bounded in $L^2(0, T; H^1(\Omega))$ may have subsequences that converge strongly in $L^2(0, T; L^2(\Omega))$.

In addition, the Aubin–Lions lemma states that if a sequence u_n is bounded in $L^2(0, T; H^1(\Omega))$ and $\partial_t u_n$ is bounded in $L^2(0, T; H^{-1}(\Omega))$, then (up to a subsequence) $u_n \rightarrow u$ strongly in $L^2(0, T; L^2(\Omega))$. These compactness results justify the strong convergence of $v_{s,n} \rightarrow v_s$ and $v_{f,n} \rightarrow v_f$ in $L^2(0, T; L^2)$, which is crucial for handling the nonlinear and interface terms in the limit.

This convergence implies the weak convergence of the time derivative:

$$\frac{d}{dt}(\mathbf{v}^{s,n}, \boldsymbol{\varphi}^s)_{0,s} \rightarrow \frac{d}{dt}(\mathbf{v}^s, \boldsymbol{\varphi}^s)_{0,s} \quad \text{in } H^{-1}(0, T),$$

for all $\boldsymbol{\varphi}^s \in V_s$ independent of time.

Moreover, we have

$$(\sigma_s(\mathbf{u}^{s,n}), \varepsilon(\boldsymbol{\varphi}))_{0,s} \rightarrow (\sigma_s(\mathbf{u}^s), \varepsilon(\boldsymbol{\varphi}))_{0,s} \quad \text{a.e. in } t,$$

for all $\boldsymbol{\varphi} \in V_s$.

On the other hand, from the uniform bound of $\mathbf{v}^{f,n}$ in $L^2(0, T; V_f)$, we can extract a subsequence $\mathbf{v}^{f,n} \rightharpoonup \mathbf{v}^f$ weakly in $L^2(0, T; V_f)$, which gives

$$(\varepsilon(\mathbf{v}^{f,n}), \varepsilon(\boldsymbol{\varphi}^f))_{0,f} \rightarrow (\varepsilon(\mathbf{v}^f), \varepsilon(\boldsymbol{\varphi}^f))_{0,f} \quad \text{a.e. in } t,$$

for all $\boldsymbol{\varphi}^f \in V_f$.

Step 3: Passing to the limit in the nonlinear term. As in [2], it remains to show that for all time-independent $\boldsymbol{\varphi}^f \in V_f$, we have

$$\int_{t_0}^t (B_n(\mathbf{v}^{f,n}) - B(\mathbf{v}^f), \boldsymbol{\varphi}^f)_{0,f} dt \rightarrow 0. \quad (4.41)$$

Let $D_n \equiv \{t \in (0, T); \|\mathbf{v}^{f,n}(t)\|_{1,f} \geq n\}$, and denote by $m(D_n)$ its Lebesgue measure.

We recall that there exists a constant $C \equiv C(\mathbf{v}_0^f, \mathbf{v}_0^s, \mathbf{f}^f, \mathbf{f}^s, \mathbf{g}_N^f, \mathbf{h}, \mathbf{g}, T) > 0$ such that

$$\int_{D_n} \|\mathbf{v}^{f,n}(s)\|_{1,f}^2 ds \leq C, \quad (4.42)$$

which implies

$$C \geq \int_{D_n} \|\mathbf{v}^{f,n}(s)\|_{1,f}^2 ds \geq m(D_n) \cdot n^2, \quad (4.43)$$

hence,

$$m(D_n) \leq \frac{C}{n^2}. \quad (4.44)$$

Step 4: Passing to the limit in the interface terms.. We now verify the convergence of the interface terms. Since $v_{f,n} \rightarrow v_f$ weakly in $L^2(0, T; V_f)$ and $v_{s,n} \rightarrow v_s$ weakly in $L^2(0, T; V_s)$, and the trace operator $H^1(\Omega) \rightarrow H^{1/2}(\Sigma)$ is continuous, it follows that

$$v_{f,n}|_{\Sigma} \rightharpoonup v_f|_{\Sigma}, \quad v_{s,n}|_{\Sigma} \rightharpoonup v_s|_{\Sigma} \quad \text{in } L^2(0, T; H^{1/2}(\Sigma)^d).$$

Hence, the jumps and averages also converge weakly in $L^2(0, T; H^{1/2}(\Sigma)^d)$. Since the matrices $M(x)$, $S(x)$ are bounded and measurable with uniformly positive eigenvalues (see ellipticity condition (1.2)), the bilinear forms

$$\langle M\overline{v_n}, \overline{\varphi} \rangle_{\Sigma}, \quad \langle S[[v_n]], [[\varphi]] \rangle_{\Sigma}$$

are continuous in the weak topology of $L^2(0, T; H^{1/2}(\Sigma)^d)$. Therefore, for any test function $\varphi \in V$, we have

$$\langle Mv_n, \varphi \rangle_{\Sigma} \rightarrow \langle Mv, \varphi \rangle_{\Sigma}, \quad \langle S[[v_n]], [[\varphi]] \rangle_{\Sigma} \rightarrow \langle S[[v]], [[\varphi]] \rangle_{\Sigma}.$$

The source interface terms $\langle h, \varphi \rangle_{\Sigma}$ and $\langle g, [[\varphi]] \rangle_{\Sigma}$ do not depend on n , and hence converge trivially. Thus, all interface contributions pass to the limit in the variational formulation.

With that established, the remainder of the proof follows exactly as in [2]. This completes the proof of the global existence of weak solutions. $\square$

Theorem 4.7 (Uniqueness of solution for $n=2$). *Let $\Omega_f, \Omega_s \subset \mathbb{R}^2$. The weak solution to the interaction problem (2.3a)–(2.3i) is unique.*

Proof. Given $(u_0^s, v_0^s, v_0^f) \in V_s \times V_s \times V_f$. Let (v_1^f, v_1^s) and (v_2^f, v_2^s) be two weak solutions of (P) with the same initial data.

We assume the existence of two solutions $(v_1^s, v_1^f), (v_2^s, v_2^f) \in V_s \times V_f$ of the weak form (4.18). Then $(v^s, v^f) = (v_2^s - v_1^s, v_2^f - v_1^f)$ satisfy, for all $\varphi^f \in V_f, \varphi^s \in V_s$, the following

$$\begin{aligned} & \left(\frac{\partial v^f}{\partial t}, \varphi^f \right)_{0,f} + \left(\frac{\partial v^s}{\partial t}, \varphi^s \right)_{0,s} + (\varepsilon(v^f), \varepsilon(\varphi^f))_{0,f} + (\sigma_s(u^s), \varepsilon(\varphi^s))_{0,s} \\ & + b(v_2^f, v_2^f, \varphi^f) - b(v_1^f, v_1^f, \varphi^f) + \langle M\overline{v}, \overline{\varphi} \rangle_{\Sigma} + \langle S[[v]], [[\varphi]] \rangle_{\Sigma} = 0 \end{aligned} \quad (4.45)$$

By using the identity $v_2^f = v_1^f + v^f$ and the bilinearity of b in its first two arguments, we manipulate the term $b(v_2^f, v_2^f, \varphi^f)$ to get

$$\begin{aligned} b(v_2^f, v_2^f, \varphi^f) &= b(v_1^f + v^f, v_1^f + v^f, \varphi^f) \\ &= b(v_1^f, v_1^f + v^f, \varphi^f) + b(v^f, v_1^f + v^f, \varphi^f) \\ &= b(v_1^f, v_1^f, \varphi^f) + b(v_1^f, v^f, \varphi^f) + b(v^f, v_2^f, \varphi^f). \end{aligned}$$

Subtracting the new expression of $b(v_2^f, v_2^f, \varphi^f)$ into (4.45), we obtain:

$$\begin{aligned} & \left(\frac{\partial v^f}{\partial t}, \varphi^f \right)_{0,f} + \left(\frac{\partial v^s}{\partial t}, \varphi^s \right)_{0,s} + (\varepsilon(v^f), \varepsilon(\varphi^f))_{0,f} + b(v_1^f, v^f, \varphi^f) + b(v^f, v_2^f, \varphi^f) \\ & + (\sigma_s(v^s), \varepsilon(\varphi^s))_{0,s} + \langle M\overline{v}, \overline{\varphi} \rangle_{\Sigma} + \langle S[[v]], [[\varphi]] \rangle_{\Sigma} = 0 \end{aligned} \quad (4.46)$$

Taking $\varphi^f = v^f$ and $\varphi^s = v^s$ and noting that $b(v_1^f, v^f, v^f) = 0$ (because $\nabla \cdot v_1^f = 0$) and dropping some positive terms yields

$$\frac{1}{2} \frac{d}{dt} \|v^f\|_{0,f}^2 + \frac{1}{2} \frac{d}{dt} \|v^s\|_{0,s}^2 + (\varepsilon(v^f), \varepsilon(v^f))_{0,f} + (\sigma_s(v^s), \varepsilon(v^s))_{0,s} \leq -b(v^f, v_2^f, v^f) \quad (4.47)$$

Now using properties of the trilinear form b in dimension $d = 2$ (see lemma 4.4), we bound the term $b(v^f, v_2^f, v^f)$ as

$$\begin{aligned} |b(v^f, v_2^f, v^f)| &\leq c_1 \|v^f\|_{1/2,f} \|v_2^f\|_{1,f} \|v^f\|_{1/2,f} + c_2 \|v^f\|_{1/2,f} \|v_2^f\|_{3/4,f} \|v^f\|_{3/4,f} \\ &\leq c_1 \|v^f\|_{0,f} \|v^f\|_{1,f} \|v_2^f\|_{1,f} + c_2 \|v^f\|_{0,f}^{3/4} \|v^f\|_{1,f}^{5/4} \|v_2^f\|_{1,f}^{3/4} \|v_2^f\|_{0,f}^{1/4} \end{aligned} \quad (4.48)$$

Applying Young's inequality gives

$$|b(v^f, v_2^f, v^f)| \leq 4c_1^2 \|v^f\|_{0,f}^2 \|v_2^f\|_{1,f}^2 + \frac{1}{2} \|v^f\|_{1,f}^2 + 4^{3/5} c_2^{8/3} \|v^f\|_{0,f}^2 \|v_2^f\|_{1,f}^2 \|v_2^f\|_{0,f}^{2/3} \quad (4.49)$$

Setting $y(t) = \|v^f(t)\|_{0,\Omega_f}^2 + \|v^s(t)\|_{0,\Omega_s}^2$, $K_1 = 4c_1^2$ and $K_2 = 4^{3/5} c_2^{8/3}$ and using the estimates above, the inequality (4.47) becomes

$$\frac{1}{2} \|v^f(t)\|_{0,\Omega_f}^2 + \frac{1}{2} \frac{d}{dt} y(t) \leq \|v^f(t)\|_{0,\Omega_f}^2 \|v_2^f(t)\|_{1,\Omega_f}^2 \left(K_1 + K_2 \|v_2^f(t)\|_{0,\Omega_f}^{2/3} \right) \quad (4.50)$$

We drop the term $\|v^f(t)\|_{0,\Omega_f}^2$ and integrate in time

$$\begin{aligned} y(t) &\leq y(0) + \int_0^t \|v^f(s)\|_{0,\Omega_f}^2 \|v_2^f(s)\|_{1,\Omega_f}^2 \left(K_1 + K_2 \|v_2^f(s)\|_{0,\Omega_f}^{2/3} \right) ds \\ &\leq y(0) + \int_0^t y(s) \|v_2^f(s)\|_{1,\Omega_f}^2 \left(K_1 + K_2 \|v_2^f(s)\|_{0,\Omega_f}^{2/3} \right) ds \end{aligned} \quad (4.51)$$

Finally, we apply Gronwall's inequality and use the boundedness of any weak solution $v_2^f \in L^2(0, T; V^f) \cup L^\infty(0, T; H)$:

$$y(t) \leq y(0) \exp \left[\left(K_1 + K_2 \|v_2^f\|_{L^\infty(0,T;H)}^{2/3} \right) \int_0^t \|v_2^f(s)\|_{1,\Omega_s}^2 ds \right] \quad (4.52)$$

However, the initial condition $y(0) = \|v^f(0, \cdot)\|_{0,\Omega_f}^2 + \|v^s(0, \cdot)\|_{0,\Omega_s}^2$ is identically null, and the right hand side of inequality (4.52) is finite. Hence, $y(t) = 0$ a.e. t , thus $(v^f, v^s) = 0$ meaning that $v_1^f = v_2^f$ and $v_1^s = v_2^s$, and uniqueness is acquired. $\square$

5. Conclusion. This work presents a generalized mathematical framework for fluid-structure interaction (FSI) problems using the fictitious domain method with parameterized interface conditions. By embedding the structure within an extended fluid domain and formulating the problem in a unified product space, the authors rigorously establish the existence of weak solutions using semigroup theory and abstract nonlinear evolution equations. The flexibility of the interface formulation allows the model to accommodate a broad range of physical scenarios, including partial slip and stress discontinuities. In two dimensions, uniqueness is proven using energy estimates and properties of the trilinear form. This contribution advances the theoretical understanding of FSI systems and opens the door to further generalizations and numerical developments.

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